Section C will contain data questions and will require students to select the necessary data from the data booklet. They will be provided with data from a laboratory experiment and asked a series of questions on it. The longer timing of the examination reflects the style of the questions for Section C.

Students will be able to show their full ability in Sections B and C as these will contain areas where they will be stretched and challenged.

Quality of written communication will be assessed in this examination, in either Section B or C. The data booklet can be used throughout the examination for this unit.

4.3 How fast? – rates

Knowledge of the concepts introduced in *Unit 2, Topic 2.8: Kinetics* will be assumed and extended in this topic.

During this topic there will be the opportunity to carry out a number of internal assessment activities. Please see *Appendix 9* for more details.

Students will be assessed on their ability to:

a demonstrate an understanding of the terms 'rate of reaction', 'rate equation', 'order of reaction', 'rate constant', 'half-life', 'rate-determining step', 'activation energy', 'heterogeneous and homogenous catalyst'

b select and describe a suitable experimental technique to obtain rate data for a given reaction, eg colorimetry, mass change and volume of gas evolved

c investigate reactions which produce data that can be used to calculate the rate of the reaction, its half-life from concentration or volume against time graphs, eg a clock reaction

d present and interpret the results of kinetic measurements in graphical form, including concentration-time and rate-concentration graphs e investigate the reaction of iodine with propanone in acid to obtain data for the order with respect to the reactants and the hydrogen ion and make predictions about molecules/ions involved in the rate-determining step and possible mechanism (details of the actual mechanism can be discussed at a later stage in this topic)

f deduce from experimental data for reactions with zero, first and second order kinetics:

  i half-life (the relationship between half-life and rate constant will be given if required)

  ii order of reaction

  iii rate equation

  iv rate-determining step related to reaction mechanisms

  v activation energy (by graphical methods only; the Arrhenius equation will be given if needed)

g investigate the activation energy of a reaction, eg oxidation of iodide ions by iodate(V)

h apply a knowledge of the rate equations for the hydrolysis of halogenoalkanes to deduce the mechanisms for primary and tertiary halogenoalkane hydrolysis and to deduce the mechanism for the reaction between propanone and iodine

i demonstrate that the mechanisms proposed for the hydrolysis of halogenoalkanes are consistent with the experimentally determined orders of reactions, and that a proposed mechanism for the reaction between propanone and iodine is consistent with the data from the experiment in 4.3e

j use kinetic data as evidence for S_N1 or S_N2 mechanisms in the nucleophilic substitution reactions of halogenoalkanes.